

VIP C2 PROGRAMME — FULL STACK DEVELOPMENT WITH
MERN

Solution Architecture

ShopEZ – MERN E-Commerce Application

Phase: Project Design Phase

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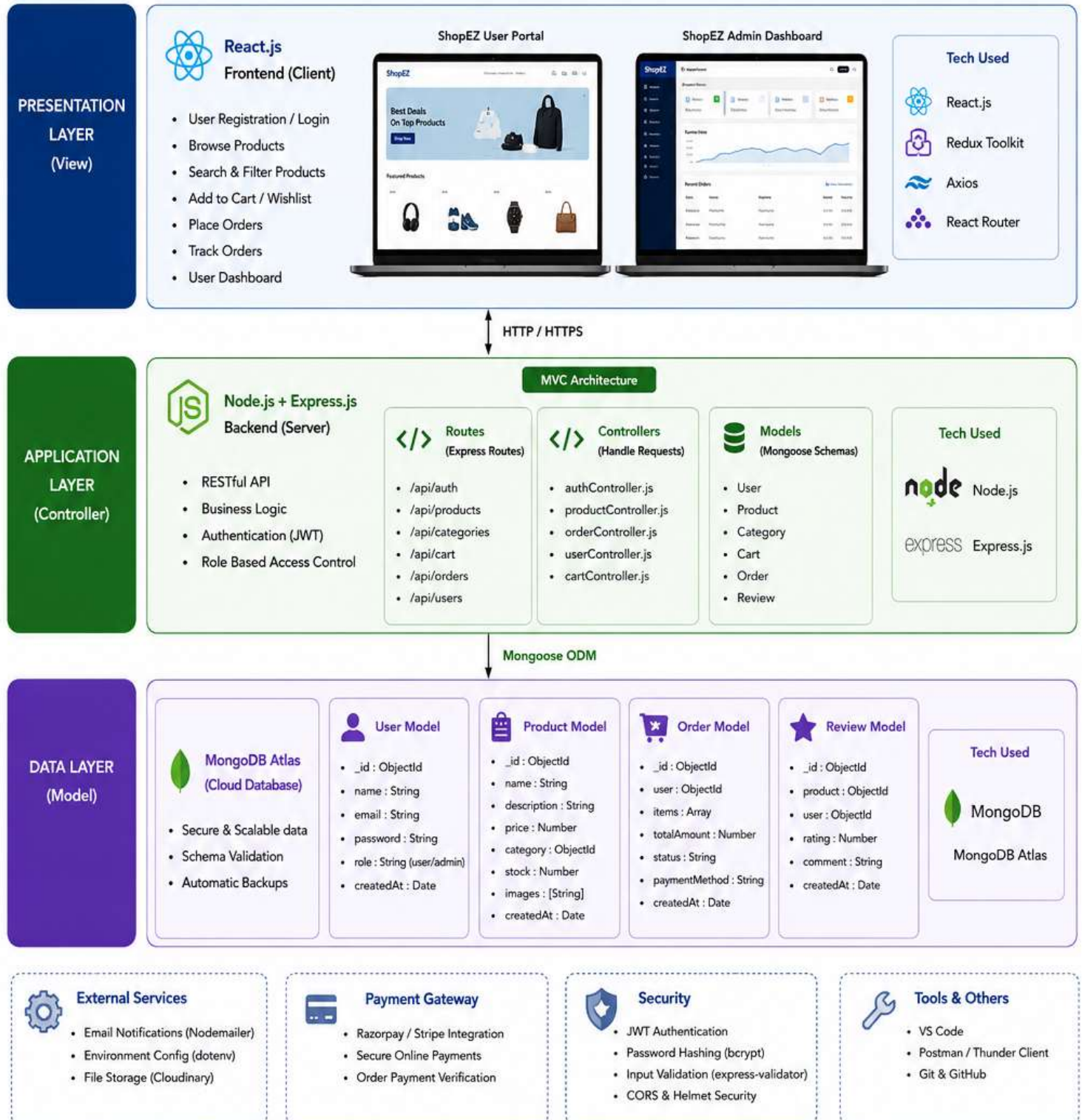
Classification: Project Documentation – For Academic Submission

Solution Architecture

1. Layered System Architecture

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ShopEZ MERN E-Commerce Application



Building a Secure, Scalable and Efficient E-Commerce Platform
using MERN Stack

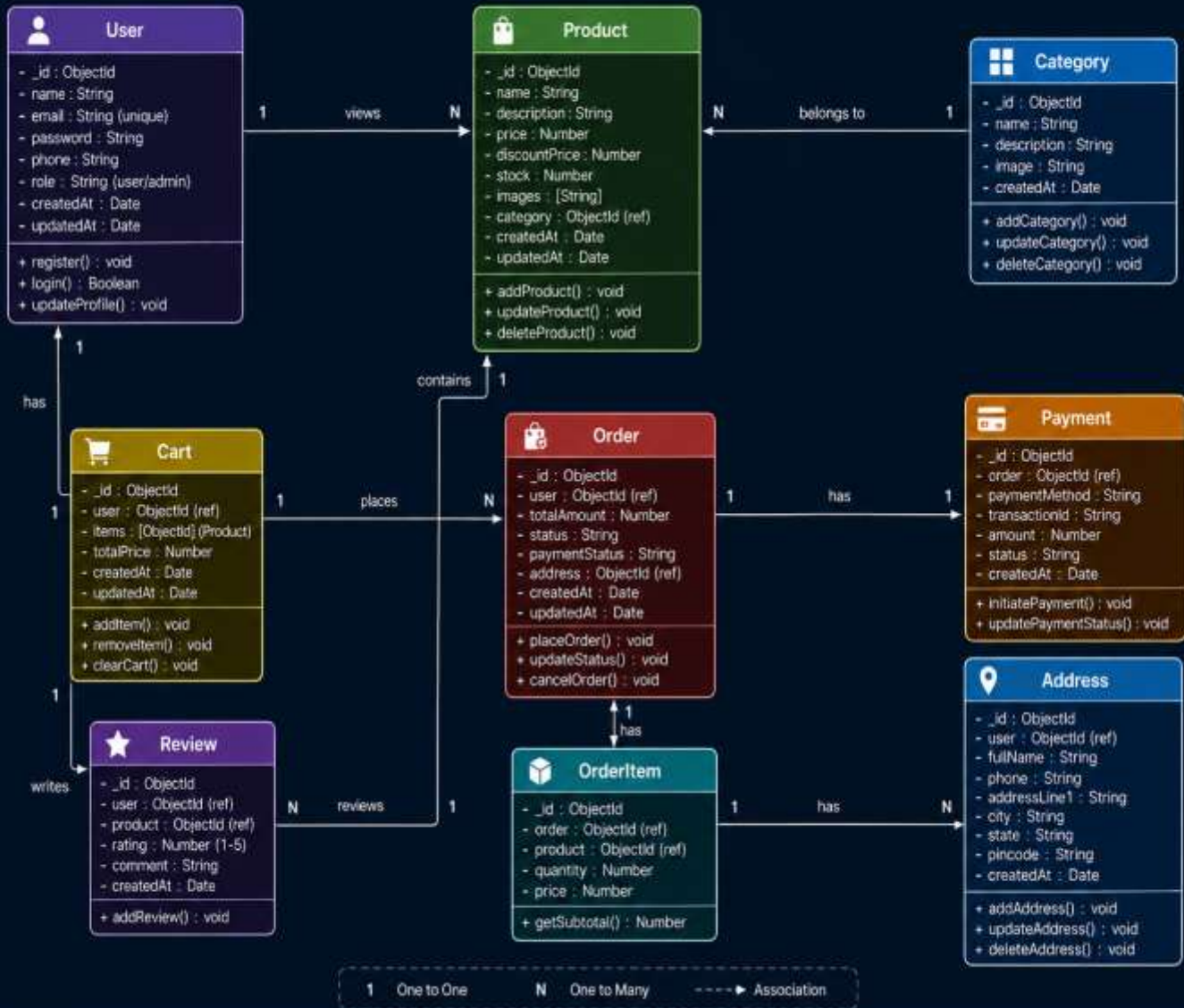


2. MVC Pattern Mapping

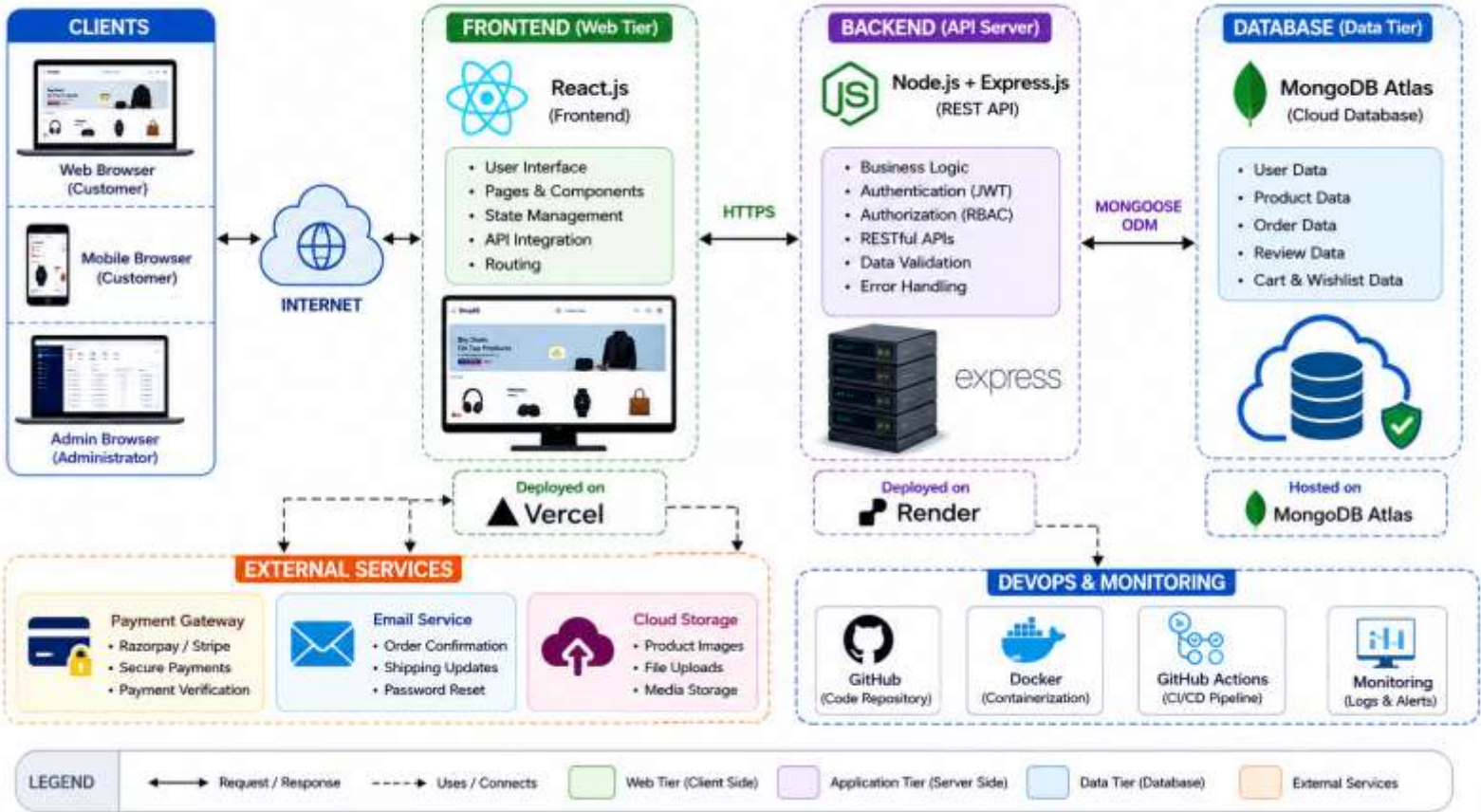
MVC Layer	MERN Component	Responsibilities
View	React.js (Frontend)	User Interface Product Catalog Shopping Cart
Controller	Node.js + Express.js	API Handling. Order Processing. Authentication
Model	MongoDB + Mongoose	Product Data. User Data. Orders & Payments.

3. UML Class Diagram — Mongoose Models

3. UML Class Diagram — ShopEZ E-Commerce Application



4. Deployment Topology — ShopEZ E-Commerce Application



5. Deployment URL Reference

Component	Platform	URL
Frontend	React	https://localhost:3000
Backend	Node.js	https://localhost:5000
Database	MongoDB Atlas	Mongodb+srv://your-cluster.mongodb.net